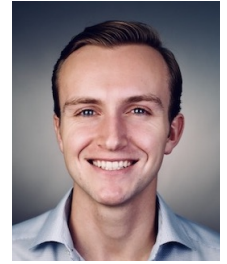


Dylan L Randle

[Website](#) • [LinkedIn](#) • [GitHub](#) • [Scholar](#)



SUMMARY

Technical lead with 8+ years experience in AI, robotics, and machine learning. Expert in developing closed-loop policies for dexterous manipulation using imitation and reinforcement learning. Proven track record building and deploying AI/ML systems for robotics, computer vision, and natural language processing.

EXPERIENCE

Amazon Robotics

North Reading, MA, USA

Senior Applied Scientist

Apr 2024 – Present

- Tech lead for closed-loop imitation and reinforcement learning systems for dexterous bimanual manipulation.
- Delivered Amazon's first bimanual manipulation policies, achieving a 92% success rate on unseen items. Technology demonstrated to Jeff Bezos, Andy Jassy, and the Board of Directors.

Senior Data Scientist

Apr 2023 – Apr 2024

- Developed production ML systems for robotic manipulation (grasp generation, damage prediction, box packing).
- Deployed first learned action model for Amazon's Sparrow robot. Delivered performance improvements of +35% and savings of \$10 million/year.

Data Scientist II

Jul 2020 – Apr 2023

- Developed ML path planning optimization system for large-scale mobile robot fleets.
- Published research demonstrating a 15% throughput increase, identifying \$150 million in annualized efficiency gains for mobile robot fleets.

Data Scientist I

Jun 2019 – Aug 2019

- Developed AutoML system for training, evaluating, and interpreting ML models trained on trillion-row robotics datasets.
- Used by multiple scientists to speed up research and analysis workflows.

Hubdoc

Toronto, ON, Canada

Data Scientist

Feb 2017 – Jul 2018

- Started and led ML team from ideation to \$70 million USD acquisition.
- Developed ML-based NLP system for automated data extraction from unstructured financial documents.
- Reduced data extraction time from hours to seconds.

BMO Capital Markets

Toronto, Canada

Financial Products Analyst

Jun 2014 – Aug 2014

- Developed interest rate swap/swaption delta-hedging optimization algorithm.
- Uncovered market opportunities for fixed-income traders.

EDUCATION

Harvard University

Cambridge, MA, USA

Master of Science in Data Science (GPA: 4.0)

Aug 2018 – May 2020

- Thesis: "Unsupervised Neural Network Methods for Solving Differential Equations".
- Recognized with Scholarship in Applied Computation and Distinction in Teaching.
- Research and coursework focused on machine learning.

University of California, Berkeley

Bachelor of Science in Industrial Engineering & Operations Research (GPA: 3.9)

Berkeley, CA, USA

Aug 2012 – May 2016

- Recognized with High Honors (*magna cum laude*) and Frank Kraft Award.
- Inducted into Phi Beta Kappa, Tau Beta Pi, Alpha Pi Mu.
- Coursework focused on statistics and optimization.

PUBLICATIONS

- **Demonstrating Multi-Suction Item Picking at Scale via Multi-Modal Learning of Pick Success.** C Wang, J van Baar, C Mitash, S Li, **D Randle**, W Wang, S Sontakke, K E Bekris, K Katyal. RSS 2025.
- **MuST: Multi-Head Skill Transformer for Long-Horizon Dexterous Manipulation with Skill Progress.** K Gao, F Wang, E Aduh, **D Randle**, J Shi. ICRA 2025.
- **Learning Object Properties Using Robot Proprioception via Differentiable Robot-Object Interaction.** PY Chen, C Liu, P Ma, J Eastman, D Rus, **D Randle**, Y Ivanov, W Matusik. ICRA 2025.
- **Avoiding Object Damage in Robotic Manipulation.** E Aduh, F Wang, **D Randle**, K Wang, P Shah, C Mitash, M Nambi. IROS 2024.
- **DEQGAN: Learning the Loss Function for PINNs with Generative Adversarial Networks.** B Bullwinkel*, **D Randle***, P Protopapas, D Sondak. ICML 2022, AI for Science. *Equal contribution.
- **Unsupervised Learning of Solutions to Differential Equations with Generative Adversarial Networks.** **D Randle**, P Protopapas, D Sondak. arXiv:2007.11133, 2020.
- **Unsupervised Neural Network Methods for Solving Differential Equations.** **D Randle**. Master's Thesis, Harvard University, 2020.

PROJECTS

- **Rubik's Cube Solving Robot** (Nov 2024): Built a robot that solves a cube in under 3 seconds using camera-based perception, Kociemba's planning algorithm, and custom stepper-motor hardware.
- **Real-Time Quantitative Trading System** (Jun 2023): Built and deployed a live quantitative trading system that uses real-time NLP analysis of news articles to generate market signals and execute trades.
- **Golf Swing Analysis with Computer Vision** (Jul 2020): Applied monocular 3D human-pose estimation to analyze body mechanics throughout a golf swing.
- **ResNet Variational Autoencoder** (Jun 2020): Trained a residual-network VAE on CelebA to learn a generative latent representation and synthesize faces.
- **Differentiable Neural Architecture Search** (Dec 2019): Partnered with Google AI to adapt and evaluate DARTS across materials science, astronomy, and medical-imaging datasets.
- **Interpretable Reinforcement Learning** (Dec 2019): Distilled black-box policies into readable decision sets using DAGger, quantified the interpretability-performance tradeoff in an HIV treatment simulator, and released `pynterp`, an open-source package providing some interpretable algorithms.
- **Distributed YouTube-8M Classification** (May 2019): Trained bidirectional LSTM video classifiers with Spark, TensorFlow, Elephas, HDFS, and a custom AWS EMR cluster.
- **Causal LSTM Microbiome Modeling** (May 2019): Used sparse LSTMs, neural Granger-causality analysis, and Bayesian optimization to compare microbial interactions in healthy and IBD mice.
- **Bayesian Generative Adversarial Networks** (Dec 2018): Reproduced and evaluated Bayesian GANs using stochastic-gradient HMC, demonstrating mode diversity on four-shapes data and semi-supervised MNIST.
- **Twitter Troll Detection** (Dec 2018): Built NLP classifiers for identifying Internet Research Agency tweets, achieving 95.9% accuracy on a temporally separated test set.
- **Automatic Differentiation from Scratch** (Dec 2018): Built and published the `dragongrad` Python package with forward- and reverse-mode differentiation, Jacobians, and gradient-based optimization.

TECHNICAL SKILLS

- **Languages:** Python, C++, Javascript/Typescript, SQL.
- **Libraries:** PyTorch, Keras/Tensorflow, OpenCV, Open3D, Pandas, NumPy, SciPy, Scikit-Learn, React.
- **Platforms:** AWS, Docker, Firebase, Linux, MacOS.